

A decorative graphic on the left side of the slide, consisting of a network of white lines and small circles on a dark blue background, resembling a circuit board or a neural network.

THE BRIDGE CROSSING PROBLEM

DATA STRUCTURES AND ALGORITHMS 2

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An abstract graphic on the left side of the slide, consisting of a network of thin, light-blue lines and small circles, resembling a circuit board or a neural network diagram. The lines are vertical and horizontal, with some diagonal connections, and the circles are placed at various points along these lines.

BRIDGE CROSSING

CAN YOU SOLVE IT??

Activity: Can you solve this problem?

n friends need to cross a bridge in the dark, but only have one flashlight. In addition, the bridge can only hold the weight of two people at a time. Given the walking speeds of each person $S = \{s_1, s_2, \dots, s_n\}$, give an algorithm that gets all n people across the bridge as quickly as possible.

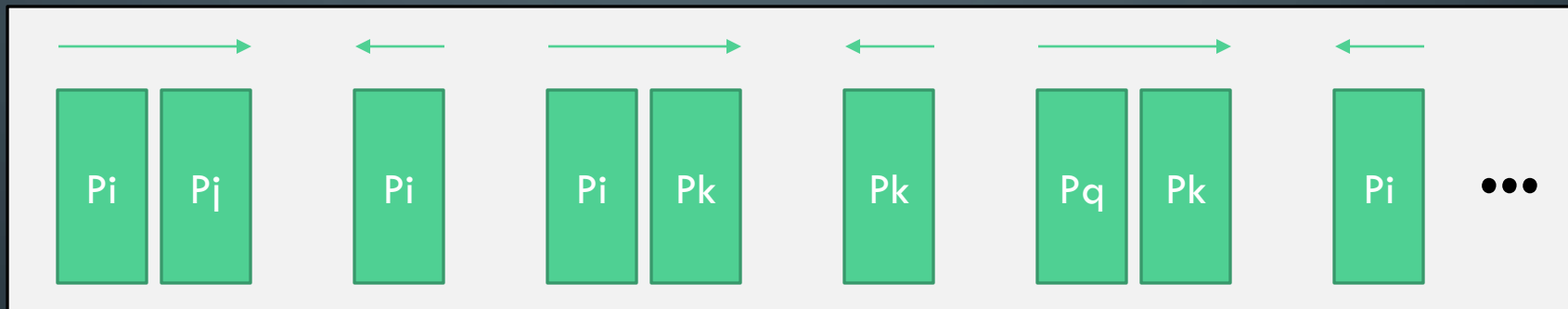
****Assume $s_1 \leq s_2 \leq \dots \leq s_n$**

****If two people cross together, they walk at the slower person's speed**

OPTIMAL SUBSTRUCTURE...

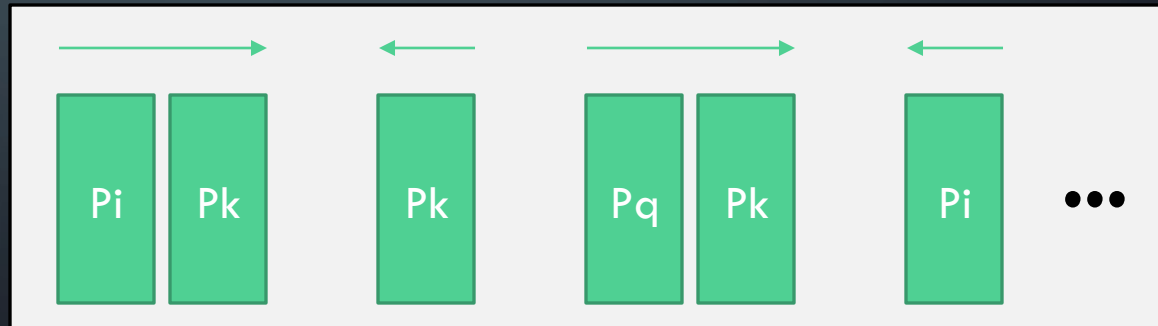
Does this problem have optimal substructure?

Consider an optimal solution $S =$:



If this is optimal overall,

then is this optimal for
“rest” of the schedule?



GREEDY CHOICE

Activity: Can you solve this problem?

n friends need to cross a bridge in the dark, but only have one flashlight. In addition, the bridge can only hold the weight of two people at a time. Given the walking speeds of each person $S = \{s_1, s_2, \dots, s_n\}$, give an algorithm that gets all n people across the bridge as quickly as possible.

*Take a few minutes and try to come up with a solution.
Let me know once you have a greedy choice*

GREEDY CHOICE!

Hint 1:

There are a lot of ways to think about the greedy choice here, but consider this: are there any people who, once across, will never come back to the original side of the bridge (you can be "done" with them)?

Hint 2:

You don't need the greedy choice to be one move across the bridge. You can greedily choose the first few moves that accomplish some sub-goal...

CAN YOU SOLVE IT??

Possible solution number 1:

s_1 (fastest) escorts s_n (slowest) across (and comes back)

Side Effects:

s_n is across bridge and will never come back

sub-problem is simply solve($\{s_1, s_2, \dots, s_{n-1}\}$)

This does NOT always work. Can you find a counter-example??

CAN YOU SOLVE IT??

Possible solution number 2:

s_1 and s_2 escort the two slowest members s_{n-1} and s_n

- s_1 and s_2 go across
- s_1 returns
- s_{n-1} and s_n cross together
- s_2 returns

...and repeat with next two slowest members.

CAN YOU SOLVE IT??

Possible solution number 2:

s_1 and s_2 escort the two slowest members s_{n-1} and s_n

Side Effects:

s_n and s_{n-1} are *BOTH* across the bridge and will never come back
sub-problem is simply $\text{solve}(\{s_1, s_2, \dots, s_{n-2}\})$

This ALSO does not always work. Counter-example?

SOLUTION

Solution: Greedy algorithm is to try to get the **two slowest people** across as quickly as possible. Then, use recursion on rest of the input

See which of the previous two techniques is better:

1. s_1 escorts

$$\text{Cost} = s_n + s_1 + s_{n-1} + s_1$$

2. s_{n-1} and s_n go together

$$\text{Cost} = s_2 + s_1 + s_n + s_2$$

SOLUTION

See which of the previous two techniques is better:

1. s_1 escorts

$$C_1 = s_n + s_1 + s_{n-1} + s_1$$

2. s_n and s_{n-1} go together

$$C_2 = s_2 + s_1 + s_n + s_2$$

Difference is:

$$\begin{aligned} C_1 - C_2 &= (s_n + s_1 + s_{n-1} + s_1) - (s_2 + s_1 + s_n + s_2) \\ &= s_{n-1} + s_1 - 2s_2 \end{aligned}$$

****If this value is positive, do approach 2, otherwise approach 1**